



Turquoise

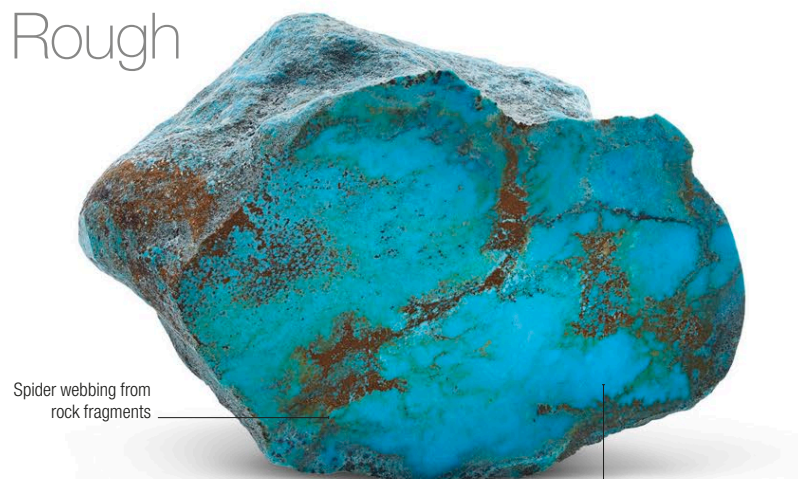
△ **Cabochon** incorporating typical spider-web markings

Beads made from turquoise dating back to c.5000 BCE have been found in Mesopotamia (present-day Iraq), making it one of the first gems to be mined and cut. It is relatively soft and easy to work and can be polished, made into beads, carved, and used for cameos. For most gem uses, however, turquoise is cut *en cabochon*. It varies in colour from sky-blue to green, depending on the amount of iron and copper it contains. Turquoise is porous and its colour may deteriorate if worn frequently close to the skin.

Specification

Chemical name Copper aluminium phosphate | **Formula** $\text{CuAl}_6(\text{PO}_4)_4(\text{OH})_8 \cdot \text{H}_2\text{O}$ | **Colours** Blue, green | **Structure** Triclinic
Hardness 5–6 | **SG** 2.4–2.9 | **RI** 1.61–1.65 | **Lustre** Waxy
Streak White-green | **Locations** Iran, China, USA, Mexico, Chile, Africa, Australia, Siberia, England, Belgium, France, Poland

Rough



Spider webbing from rock fragments

Prized pure colour

Bisbee rough | This fine example from Bisbee, Arizona, US, shows the “spider webbing” (dark veining) typical of blue turquoise from that locality.

Matrix (host rock)

Thin layering



Turquoise rough | A piece of massive turquoise like this, with thin layers of the mineral interlayered with matrix, needs skilful cutting to extract the gem.

Cut



Spider webbing

Pendeloque cabochon | The high-domed, tear-drop shaped cabochon cut of this classic Bisbee turquoise complements the gem’s attractive black spider webbing.

Persian blue

Top of the range

Turquoise from Nishapur, Iran (formerly Persia), is considered by many to be the finest quality and has been mined for centuries. This turquoise, usually referred to as “Persian”, tends to be harder and of a more even colour than North American turquoise, and it is always sky-blue, never green. Turquoise has embellished thrones, sword hilts, horse trappings, daggers, bowls, cups, and other ornaments over the centuries, as well as being used extensively in jewellery.



Persian ornament This ornament is engraved and inlaid with gold in the highest expression of the lapidary art.

Iron oxides



Bisbee sample | This oval stone, naturally patterned with iron oxides, comes from the same type of turquoise as the Bisbee rough (above left).

Colour looks “muddy”



Imitation stone | This oval cabochon is cut from synthetic turquoise. It lacks the colour and texture of natural turquoise, though its uniform hue can be an advantage.